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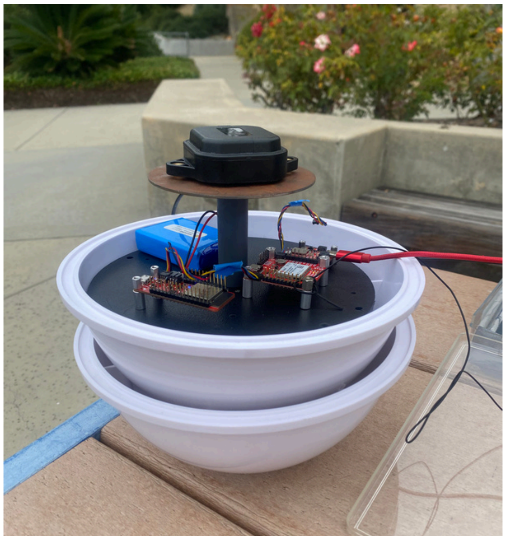
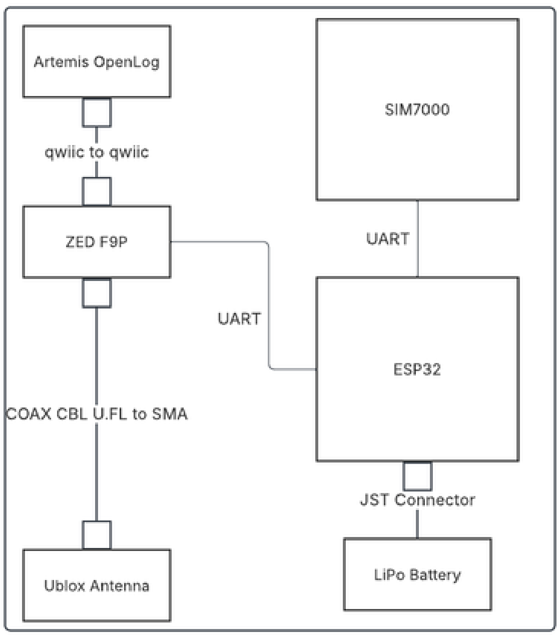
SCRIPP MARINE PHYSICAL LAB - MARINE RTK BUOY SYSTEM





What?

- Designed and built an autonomous marine buoy GPS system for oceanographic research
- Achieve centimeter-level positioning accuracy in the field

How?

- Integrated 4 hardware components (ESP32, u-blox ZED-F9P, SIM7000 LTE modem, OpenLog Artemis IMU) communicating over UART and I2C simultaneously
- Wrote C++ firmware to manage cellular connectivity, NTRIP correction streaming, and graceful field shutdown
- Validated results using Python (pyubx2) and MATLAB over a 7.5-hour overnight test

Results

- 0.97 cm horizontal precision
- 94% RTK fix rate over 265,000+ position points
- 7.5-hour autonomous field operation